

Punctuation Restoration for Kyrgyz Language: A Comparative Study of Multilingual Transformer Models

ANONYMOUS AUTHOR(S)

We present the first punctuation restoration system for Kyrgyz, a low-resource agglutinative language of the Turkic family with approximately 7 million speakers. Punctuation restoration is a critical post-processing step for automatic speech recognition (ASR) systems, yet no prior work has addressed this task for Kyrgyz. We construct a labeled dataset of 16,028 sentences drawn from literary texts, Wikipedia, and news sources, including a targeted question-mark expansion that raises the QUESTION token count from 940 to 2,938. We compare three transformer encoders fine-tuned for token-level classification over the labels 0, PERIOD, COMMA, and QUESTION: two multilingual models (mBERT, XLM-RoBERTa-base) and the Kyrgyz-specific monolingual KyrgyzBERT. XLM-RoBERTa achieves the best overall weighted F1-score of 94.4%, outperforming mBERT (93.0%), KyrgyzBERT (92.1%), and a rule-based baseline (76.8%). We additionally (i) enumerate the full punctuation inventory of written Kyrgyz and justify the scope of marks covered by our system, (ii) benchmark the inference efficiency of the three transformer models, and (iii) evaluate an ASR-integration pipeline on Common Voice Kyrgyz in which our post-processor adds 589 commas and raises sentence-terminal coverage from 88% to 100% on 200 utterances produced by a Kyrgyz-fine-tuned Whisper model. We release our dataset and all three fine-tuned models publicly to support further research on Turkic language processing.¹

CCS Concepts: • **Applied computing** → **Document analysis**; • **Computing methodologies** → **Natural language processing**.

Additional Key Words and Phrases: punctuation restoration, Kyrgyz language, XLM-RoBERTa, mBERT, low-resource NLP, token classification, Turkic languages

1 Introduction

Punctuation restoration, the task of inserting appropriate punctuation marks into unpunctuated text, is an essential post-processing step for automatic speech recognition (ASR) systems. ASR output typically lacks punctuation, which degrades readability for human readers and reduces the effectiveness of downstream NLP tasks such as machine translation, summarization, and named-entity recognition [17, 26].

Early approaches to punctuation restoration relied on hand-crafted rules and n-gram language models. The introduction of recurrent neural networks (RNNs) and long short-term memory (LSTM) networks brought substantial improvements, enabling models to capture longer-range dependencies in text [11]. The transformer architecture [23] and pre-trained language models such as BERT [6] and RoBERTa [13] further advanced the state of the art, with XLM-RoBERTa [5] extending these capabilities to over 100 languages through cross-lingual pre-training on 2.5 TB of filtered CommonCrawl data.

Despite these advances, punctuation-restoration research has focused predominantly on high-resource languages such as English, Chinese, and European languages [1, 9, 16], leaving low-resource languages largely unstudied. This gap is particularly problematic for languages such as Kyrgyz, where ASR systems are beginning to emerge but lack the post-processing pipeline needed for practical deployment.

Kyrgyz presents specific challenges for punctuation restoration. As an agglutinative Turkic language with SOV (Subject-Object-Verb) word order, Kyrgyz exhibits complex morphological patterns in which a single word may carry multiple suffixes encoding grammatical relations. Converb suffixes (e.g., *-p/-yp/-ip*) frequently mark clause boundaries where commas should appear, and the verb-final structure means that sentence-ending punctuation depends on recognizing

¹Code, dataset, and model checkpoints will be released under permissive licenses upon acceptance; URLs and DOIs are withheld during double-anonymous review to preserve author anonymity.

finite verb forms. These morphosyntactic properties are not shared by the high-resource languages on which most punctuation restoration systems are evaluated.

In this paper, we make the following contributions:

- (1) We present the **first punctuation restoration system** for the Kyrgyz language, demonstrating that multilingual transfer learning is effective for agglutinative low-resource languages.
- (2) We construct and release a **labeled Kyrgyz dataset** of 16,028 sentences for punctuation restoration, including a targeted expansion that triples the QUESTION-token count to address pronounced class imbalance on interrogatives.
- (3) We provide a **three-way comparative evaluation** of two multilingual transformer models (mBERT, XLM-RoBERTa) and the Kyrgyz-specific monolingual model KyrgyzBERT, plus a rule-based baseline, with detailed per-class analysis.
- (4) We document the **complete punctuation inventory** of written Kyrgyz and explicitly justify the scope of marks our system handles (Section 2.3).
- (5) We benchmark the **inference efficiency** of the three transformer models (Section 5.5) and evaluate an **ASR-integration pipeline** on Common Voice Kyrgyz (Section 5.4), showing that even a Kyrgyz-fine-tuned Whisper fails to insert commas and that our post-processor is the component that yields a reader-usable transcript.
- (6) We present a **linguistically informed error analysis** that connects model errors to specific morphosyntactic properties of Kyrgyz.

2 Kyrgyz Language Background

Kyrgyz (ISO 639-3: *kir*) is a Turkic language of the Kipchak branch, spoken by approximately 7 million people, primarily in the Kyrgyz Republic. It is written in Cyrillic script with 36 letters, including three letters absent from the Russian Cyrillic alphabet: *Ө*, *У*, and *Ҥ*. A recent survey of Kyrgyz NLP resources [2] characterizes Kyrgyz as “severely under-resourced,” highlighting the scarcity of annotated datasets and language-specific models. Understanding the linguistic properties of Kyrgyz is essential for interpreting our model’s behavior and error patterns.

2.1 Morphological Properties

Kyrgyz is a highly agglutinative language in which words are formed by concatenating multiple suffixes to a root. A single word can encode information that would require an entire phrase in English. For example:

okutuchularybyzdyng = *oku* (read) + *-tuuchu* (agent) + *-lar* (plural) + *-ybyz* (1st person plural possessive)
+ *-dyng* (genitive) = ‘of our teachers’

This agglutinative structure has direct consequences for punctuation restoration. First, subword tokenization algorithms such as BPE (used by XLM-RoBERTa) fragment Kyrgyz words into multiple tokens, meaning the model must learn to associate punctuation decisions with token sequences rather than individual words. Second, certain morphological markers are strong indicators of punctuation placement: converb suffixes (*-p*, *-yp*, *-ip*) typically signal comma positions, as they mark clause boundaries in complex sentences.

Table 1. Punctuation marks in written Kyrgyz with their frequency in our corpus (16,028 sentences; 144,564 tokens) and coverage by our model. Sentence-final percentages are of total sentences. “Yes” and “No” indicate whether the mark is predicted by our model.

Mark	Function	Count	Sent.-final	Modeled?
,	clause/list separator	16,518	—	Yes
.	declarative sentence terminator	14,041	87.5%	Yes
?	interrogative sentence terminator	2,952	12.5%	Yes
–	dash (copula / direct speech)	3,285	—	No
!	exclamation	1,370	<0.1% ^a	No
« »	primary quotation marks	1,635	—	No
:	colon (explanation / list intro)	745	—	No
" "	secondary quotation marks	683	—	No
()	parentheses (editorial insertion)	405	—	No
–	em-dash (aside / emphasis)	170	—	No
;	semicolon	95	—	No
...	ellipsis	9	—	No

^aExclamation occurs primarily inside direct-speech constructions (e.g., “—*Ket!* — *деду*”), so it is rarely the absolute last character of a written sentence in literary corpora.

2.2 Syntactic Properties

Kyrgyz follows SOV (Subject-Object-Verb) word order, with the finite verb appearing at the end of the sentence. Period placement therefore depends on recognizing sentence-final verb forms, a signal very different from that of SVO languages such as English, where sentence boundaries are marked by subject-initial patterns.

Complex sentences in Kyrgyz are constructed using participial and converbal forms rather than relative pronouns. Subordinate clauses are typically formed by suffixing the verb, which creates long, multi-clause sentences in which comma placement depends on morphological analysis rather than on syntactic keywords.

2.3 Punctuation Inventory of Written Kyrgyz

Written Kyrgyz shares the punctuation system of modern Russian² with two minor adaptations: the dash (–) is used extensively as a copula substitute (Kyrgyz lacks an overt “to be” in the present tense), and the vocative comma appears in direct-address constructions. Table 1 lists every punctuation mark attested in our corpus, its function, and whether it is modeled by our system.

Scope of this work. We model the three punctuation marks (., , , ?) that (i) carry the highest information load for readability of ASR output and (ii) are unambiguously local: their placement is determined by the token they follow. These three marks alone cover 100% of sentence-final positions and 98.6% of all non-paired punctuation tokens in the corpus.

We explicitly *exclude* the remaining marks for the following reasons. The **exclamation mark** (!) almost never appears as the absolute sentence terminator in written Kyrgyz; it is embedded inside direct-speech segments that themselves end with a period following the closing quote or dash, and treating it as a token-level label would require additional direct-speech modeling. The **dash** (–) as a copula substitute has a placement that depends on the underlying syntactic parse (“*Мен – студентмин*” = “I am a student”), not on morphological cues, and accurate dash restoration would require

²Both languages use Cyrillic script and have been in contact for over a century; Kyrgyz orthographic and punctuation norms were standardized during the Soviet period largely following Russian conventions.

Table 2. Comparison of Kyrgyz and Turkish.

Property	Kyrgyz	Turkish
Branch	Kipchak	Oghuz
Script	Cyrillic	Latin
Speakers	~7M	~80–90M
Word order	SOV	SOV
Morphology	Agglutinative	Agglutinative
Prior research	None (this work)	Kurt & Çayır, 2023
NLP resources	Very limited	Moderate

a dependency parser, which itself does not exist for Kyrgyz at the time of writing. **Quotation marks** («», " ") and **parentheses** (()) are *paired* marks; token-level classification cannot guarantee bracket balance without a structured decoder. Finally, the **colon** (:), **semicolon** (;), **em-dash** (–), and **ellipsis** (...) are each below 1% of all punctuation tokens; training a reliable classifier for them would require either focal loss [12] or targeted data collection beyond the scope of this paper. We revisit these exclusions in Section 6.4.

2.4 Comparison with Related Languages

Table 2 compares Kyrgyz with Turkish, another Turkic language for which punctuation restoration has been studied [10]. Despite their shared Turkic heritage, the languages differ in script, branch, and available NLP resources.

3 Related Work

Transformer-based punctuation restoration. Alam et al. [1] fine-tuned XLM-RoBERTa for punctuation restoration on English and Bangla, achieving F1-scores of 87.0% on clean English text and 69.5% on Bangla. Their work demonstrated that multilingual pre-trained models can transfer to low-resource languages, though with a significant performance gap. Yi and Tao [26] combined self-attention with acoustic embeddings for English ASR transcripts. Guerreiro et al. [9] evaluated multiple transformer models across European languages, confirming that XLM-RoBERTa achieves strong cross-lingual performance even with minimal target-language data.

Multilingual and low-resource approaches. Li and Lin [11] trained a multilingual LSTM model with BPE encoding across 43 languages, showing that shared multilingual representations benefit low-resource targets. Chordia [3] compared mBERT and XLM-RoBERTa across multiple languages including Hindi and Tamil, finding that XLM-RoBERTa consistently outperformed mBERT for morphologically complex languages. Pan et al. [18] applied pre-trained models to multilingual ASR outputs, addressing domain mismatch between written training data and spoken language.

Turkic languages. Kurt and Çayır [10] developed transformer-based models for Turkish punctuation restoration, reporting macro F1-scores up to 83.9%. More recently, Görmez et al. [8] constructed a new Turkish dataset and trained CNN and transformer-encoder models, achieving an overall F-score of 90.1%. Both studies confirm the challenges of comma prediction in agglutinative Turkic languages. Kyrgyz shares agglutinative morphology and SOV word order with Turkish, so these works serve as natural reference points, despite the two languages belonging to different Turkic branches (Oghuz and Kipchak).

Streaming and real-time approaches. Poláček et al. [19] proposed a lightweight ELECTRA-based online punctuation restoration module for streaming ASR, achieving F1 of 71.2% on Czech radio transcripts with a latency of only

three words. Their work highlights the practical constraints of real-time deployment, a direction relevant for future Kyrgyz ASR systems.

Data augmentation for low-resource NLP. Wei and Zou [24] proposed Easy Data Augmentation (EDA) techniques that improve text classification performance in low-resource settings. Fu et al. [7] showed that external data selection using n-gram language models can improve punctuation restoration for speech transcripts by 1.12% F1.

Recent Kyrgyz NLP resources. Concurrent with our work, three recent resources have substantially improved the Kyrgyz NLP landscape. KyrgyzBERT [15] is a compact monolingual BERT pre-trained exclusively on Kyrgyz; we evaluate it as a baseline (Sections 5.1 and 5.5). KyrText [4] is a multi-domain 680.5M-word Kyrgyz corpus that supports continued pre-training of mBERT, XLM-R, and from-scratch RoBERTa training; a re-run of our experiments on top of these stronger Kyrgyz-adapted checkpoints is an obvious next step. Metinov [14] provides strong Russian $\leftrightarrow$ Kyrgyz MT baselines that may be useful for cross-lingual transfer experiments in future work. For Hebrew punctuation, the verbit/hebrew_punctuation Hugging Face model card provides an AlephBERT-based fine-tuned system without accompanying publication; it serves as the real-world deployment analogue that motivates our Kyrgyz-specific pipeline.

4 Methodology

4.1 Task Definition

We formulate punctuation restoration as a token-level classification problem. Given a sequence of input tokens $\mathbf{x} = (x_1, x_2, \dots, x_n)$, the model predicts a label $y_i \in \{0, \text{PERIOD}, \text{COMMA}, \text{QUESTION}\}$ for each token x_i , where the label indicates the punctuation mark that should follow the token (or 0 for no punctuation):

$$\hat{y}_i = \arg \max \text{softmax}(\mathbf{W} \cdot \mathbf{h}_i + \mathbf{b}) \quad (1)$$

where $\mathbf{h}_i \in \mathbb{R}^{768}$ is the hidden representation of token x_i from the final transformer layer, $\mathbf{W} \in \mathbb{R}^{4 \times 768}$ is a learnable weight matrix, and $\mathbf{b} \in \mathbb{R}^4$ is a bias vector.

4.2 Dataset

No punctuation restoration dataset existed for Kyrgyz prior to this work. We constructed a dataset from three sources:

- **Literary texts** (67.5%): Books in various genres (literature, science, history) downloaded from a public Kyrgyz academic library (institution name withheld for anonymous review) and converted from PDF to text.
- **Wikipedia** (20.0%): Articles from the Kyrgyz-language Wikipedia.
- **News articles** (12.5%): Texts from Kyrgyz-language news portals.

Text preprocessing proceeds in four stages. (i) **Sentence segmentation.** Raw text is split into sentences using a regular-expression pattern that matches sentence-terminal punctuation (., ?, !) followed by whitespace and an upper-case letter. We rely on a regex rather than a general-purpose sentence splitter (e.g. NLTK punkt or spaCy) because Kyrgyz is not supported by those libraries, and the regex-based splitter is the standard approach for Cyrillic low-resource languages lacking dedicated segmenters. (ii) **Word tokenization.** Each sentence is split on whitespace into whitespace-delimited tokens; subword tokenization is performed later inside the model’s tokenizer. (iii) **Label assignment.** Each token is assigned a label based on the punctuation mark immediately following it (. $\rightarrow$ PERIOD, , $\rightarrow$ COMMA, ? $\rightarrow$ QUESTION, otherwise 0); the trailing punctuation character is then stripped from the token. (iv) **Filtering.** We removed duplicate sentences, corrected encoding errors specific to Cyrillic text, and discarded sentences containing fewer than 2 tokens. Input

Table 3. Dataset statistics. Original counts refer to the initial 14,028-sentence corpus; final counts are after the targeted question-mark expansion described in Section 4.2.1.

Statistic	Original	Final (augmented)
Total sentences	14,028	16,028
Total tokens	141,626	144,564
Avg. sentence length	10.1 tokens	9.0 tokens
Period count (PERIOD)	13,564 (9.6%)	13,730 (9.5%)
Comma count (COMMA)	15,257 (10.8%)	16,437 (11.4%)
Question count (QUESTION)	940 (0.6%)	2,938 (2.0%)
No punctuation (0)	111,865 (79.0%)	111,459 (77.1%)

sequences longer than 256 subword tokens were truncated during the model’s own tokenizer step. The full preprocessing code is released with the dataset.

We split the dataset into training (76.5%), validation (8.5%), and test (15%) sets using random sampling with a fixed seed (`random_state=42`) for reproducibility.

4.2.1 Question-mark Augmentation. The initial 14,028-sentence dataset contains only 940 QUESTION tokens across all splits. After the standard 76.5/8.5/15 train/val/test partition, the test set retains only 125 questions, which is insufficient for reliable class-wise evaluation. We therefore expand the training data with 2,000 additional question-ending sentences drawn from an independent in-house cleaned Kyrgyz text corpus (a 1.6M-line UTF-8 plain-text collection of Kyrgyz literary, news, and Wikipedia material curated by the authors during prior data-collection work, with the same domain mix as the primary dataset) and deduplicated string-wise against the original dataset. Candidate sentences are required to (i) end with ?, (ii) contain 3–30 words, (iii) consist of a single sentence, and (iv) contain Cyrillic characters. Sampling uses seed 42 for reproducibility, and the resulting `question_data.json` file is released alongside the augmented training set on the dataset Hugging Face repository.

The expansion raises the QUESTION token count from 940 to 2,938 (a $3.1 \times$ increase) and the QUESTION test-set support from 125 to 470, while leaving the distribution of the other three classes largely unchanged. All experiments reported in Sections 5 onwards are conducted on this augmented dataset with an identical 76.5/8.5/15 split. The code that produces the expansion is released alongside the models.

4.3 Models

We compare three pre-trained transformer models, all fine-tuned on our augmented Kyrgyz dataset with a linear classification head appended on top of the final hidden states for token-level prediction.

mBERT (bert-base-multilingual-cased [6]) is pre-trained on Wikipedia text in 104 languages using masked language modeling and next-sentence prediction. It uses WordPiece tokenization and contains approximately 177 M parameters (12 layers, 12 attention heads, 768-dimensional hidden states).

XLM-RoBERTa-base [5] is pre-trained on 2.5 TB of filtered CommonCrawl data in 100 languages including Kyrgyz. It uses SentencePiece BPE tokenization and contains approximately 277 M parameters (12 layers, 12 attention heads, 768-dimensional hidden states). Unlike mBERT, XLM-RoBERTa was trained with larger batches and more data, and does not use next-sentence prediction.

Table 4. Training hyperparameters (identical across all three fine-tuned models).

Parameter	Value
Max sequence length	256
Batch size	16
Learning rate	5e-5
Optimizer	AdamW
Weight decay	0.01
Warmup ratio	0.1
Epochs	5
Loss function	Cross-entropy
Precision	FP16
Hardware	NVIDIA RTX 5080 (16 GB)

KyrgyzBERT [15] is a recent Kyrgyz-specific monolingual model (HuggingFace ID `metinovadilet/KyrgyzBert`). It is a compact BERT with 6 transformer layers, 8 attention heads, and 512-dimensional hidden states (approximately 35.6 M parameters); it uses WordPiece tokenization and was pre-trained exclusively on Kyrgyz text. A plausible intuition is that a model pre-trained on the target language should outperform multilingual alternatives on downstream Kyrgyz tasks; we test this hypothesis directly.

For reference, we also evaluate a **rule-based baseline**: a deterministic system that inserts a period at sentence boundaries and a comma every M tokens, where M is set to the mean clause length in the training data ($M = 5$). The baseline does not predict question marks, assigning all sentence-final positions the PERIOD label. All training and evaluation was performed using the HuggingFace Transformers library [25].

4.4 Training Configuration

All three models use the hyperparameters listed in Table 4, ensuring a fair comparison. We train for 5 epochs and select the checkpoint with the highest validation weighted F1-score.

4.5 Label Assignment

All three tokenizers split Kyrgyz words into multiple subword tokens, so each word’s punctuation label must be assigned to exactly one subtoken. We assign the label to the **last subtoken** of each word; all other subtokens receive the special ignore index -100 and are excluded from the loss computation. This strategy forces the model to accumulate contextual information across the full subword sequence before making a prediction, which is particularly relevant for agglutinative Kyrgyz words where the punctuation-relevant morphological suffixes occur at the word end.

4.6 Evaluation Metrics

We evaluate using precision (P), recall (R), and F1-score at the token level, computed both per-class and as a weighted average. We also report the full confusion matrix to analyze error patterns across punctuation classes.

Table 5. Overall punctuation restoration results (weighted average) on the augmented test set.

Model	Precision	Recall	F1
Rule-based baseline	0.773	0.765	0.768
KyrgyzBERT [15]	0.920	0.923	0.921
mBERT	0.929	0.932	0.930
XLM-RoBERTa-base	0.943	0.944	0.944

5 Experiments and Results

5.1 Main Results

Table 5 presents the overall performance of each model on the held-out test set (15% of the augmented dataset, 24,179 tokens). The per-class support is 0: 19,085, COMMA: 2,657, PERIOD: 1,967, QUESTION: 470 in the test set, with 87,412 / 12,044 / 10,009 / 2,222 tokens respectively in the training split. The training-split support of 2,222 QUESTION tokens, versus 705 in the initial 14,028-sentence corpus, is what enables the per-class gains discussed in Section 4.2.1.

XLM-RoBERTa remains the best-performing model on the augmented test set (weighted F1 0.944), followed by mBERT (0.930) and KyrgyzBERT (0.921). Despite being pre-trained exclusively on Kyrgyz, KyrgyzBERT does *not* surpass the multilingual models: its smaller capacity (6 layers, 35.6 M parameters) and smaller pre-training corpus appear to outweigh the advantage of monolingual specialization in our setting. The multilingual pre-training scale of XLM-RoBERTa, which includes Kyrgyz among 100 languages drawn from 2.5 TB of CommonCrawl, is currently a stronger basis for this task. KyrgyzBERT nonetheless has a substantial efficiency advantage, which we quantify in Section 5.5.

Zero-shot KyrgyzBERT (no fine-tuning). To quantify the contribution of task-specific fine-tuning, we additionally evaluated KyrgyzBERT in a *zero-shot* setting. We use the pre-trained masked-LM checkpoint as-is, with no exposure to our punctuation labels. For each test sentence we insert a [MASK] token after every word and, in a single forward pass, read off the softmax distribution at each mask; for every gap we compare the probabilities of ‘.’, ‘,’ and ‘?’ to the aggregate probability of all other (non-punctuation) tokens, and predict the argmax. The resulting weighted F1 is 0.769 (vs. 0.921 for the fine-tuned checkpoint), with a striking per-class breakdown: 0 F1=0.904, COMMA F1=0.505, PERIOD F1=0.001, QUESTION F1=0.000. The zero-shot model never predicts a sentence-terminal mark: across the 1,967 true PERIOD tokens in the test set it assigns PERIOD to exactly one, and it never predicts QUESTION. The cause is distributional: commas are the most frequent punctuation token in the pre-training corpus and therefore dominate the mid-sentence [MASK] distribution; without task supervision, the MLM has no way of knowing that the current [MASK] is in a sentence-terminal position. This is a strong empirical confirmation that fine-tuning on labeled punctuation data is necessary even for a Kyrgyz-specific MLM; pre-training distribution alone is not enough.

5.2 Per-Class Comparison

Table 6 presents the full per-class precision, recall, and F1-score for the three fine-tuned models on the augmented test set. XLM-RoBERTa outperforms both alternatives on three of four classes, with the largest gains on COMMA (+10.7% F1 over KyrgyzBERT, +6.9% F1 over mBERT) and QUESTION (+9.2% F1 over KyrgyzBERT, +2.1% over mBERT).

Several patterns emerge from the per-class comparison:

Table 6. Detailed per-class results for mBERT, XLM-RoBERTa, and KyrgyzBERT on the Kyrgyz augmented test set. Best F1 per class is in bold.

Class	mBERT			XLM-RoBERTa			KyrgyzBERT			Support
	P	R	F1	P	R	F1	P	R	F1	
O (no punctuation)	0.952	0.969	0.961	0.965	0.971	0.968	0.949	0.965	0.957	19,085
COMMA	0.755	0.649	0.698	0.788	0.747	0.767	0.707	0.619	0.660	2,657
PERIOD	0.960	0.980	0.970	0.965	0.979	0.972	0.966	0.976	0.971	1,967
QUESTION	0.817	0.800	0.809	0.847	0.813	0.830	0.739	0.736	0.738	470
Weighted avg.	0.929	0.932	0.930	0.943	0.944	0.944	0.920	0.923	0.921	24,179

Table 7. Comparison with punctuation restoration results for other languages. Direct comparison is limited due to differences in datasets and evaluation protocols. [†]Macro F1. [‡]Overall F-score.

Language	Model	F1	Resources	Source
English (clean)	XLM-R	0.870	High	[1]
Turkish	Transformer enc.	0.901 [‡]	Medium	[8]
Turkish	Transformers	0.839 [†]	Medium	[10]
Hungarian	BERT	0.820	Medium	[16]
Arabic	XLM-R	0.785 [†]	Medium	[21]
Kyrgyz	XLM-R	0.944	Low	This work
Bangla	XLM-R	0.695	Low	[1]
Multilingual (43 lang., incl. Hebrew, Russian)	BPE+LSTM	0.802 [‡] (news) / 0.735 [‡] (ASR)	Mixed	[11]

- PERIOD achieves near-perfect performance for all three models (F1: 0.970–0.972). Sentence boundaries in Kyrgyz are signaled by strong lexical cues (sentence-final verb forms), which even the compact KyrgyzBERT captures effectively.
- COMMA is the most challenging class across all models (XLM-R F1: 0.767, mBERT F1: 0.698, KyrgyzBERT F1: 0.660). The XLM-R advantage is driven largely by higher precision (0.788 vs. 0.707 for KyrgyzBERT and 0.755 for mBERT). The residual difficulty reflects the inherent ambiguity of comma placement at Kyrgyz converb clause boundaries.
- QUESTION benefits most from the expanded training data (Section 4.2.1): F1 improves from 0.643/0.704/0.410 on the initial 14,028-sentence corpus to 0.809/0.830/0.738 on the augmented corpus for mBERT, XLM-R, and KyrgyzBERT respectively. The largest gain (+32.8 points) is observed for KyrgyzBERT, which has the smallest pre-training corpus and therefore benefits most from additional in-class supervision.
- O (no punctuation) remains a very easy class for all three models; XLM-R again leads at F1 0.968.

Figure 1 presents the normalized confusion matrices for mBERT and XLM-RoBERTa, illustrating the error distribution across punctuation classes.

5.3 Cross-Lingual Comparison

Table 7 places our results in the context of punctuation restoration research on other languages.

Our best Kyrgyz model achieves a higher F1 (0.944) than reported for both Bangla (0.695) and English clean text (0.870). Direct comparison is limited: our dataset consists of formal written text, whereas Alam et al. [1] also evaluated

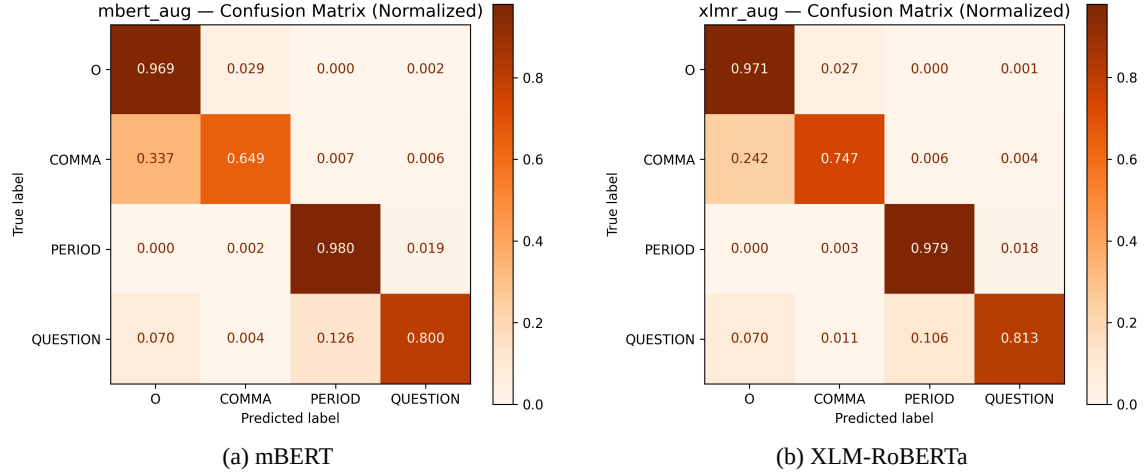


Fig. 1. Normalized confusion matrices for (a) mBERT and (b) XLM-RoBERTa on the augmented Kyrgyz test set. Values represent the proportion of each true class predicted as each label. Both models show strong diagonal dominance; XLM-RoBERTa exhibits fewer off-diagonal errors, particularly for COMMA and QUESTION.

on noisy ASR-like transcripts. The strong performance likely reflects the formal, well-structured nature of our sources (literary texts, Wikipedia), as well as the effectiveness of fine-tuning on language-specific data.

Two recent cross-lingual reference points are worth noting. (i) **Arabic:** Sakr and Torki [21] report macro F1 0.785 for an XLM-RoBERTa punctuation-restoration model on the AraPunc corpus (a six-class task over Tashkeela-derived news text). (ii) **Hebrew and Russian:** a strictly comparable per-language F1 is, to our knowledge, not currently available in the literature: the most directly relevant published benchmark is Li and Lin [11], whose multilingual model covers 43 languages including Hebrew and Russian and reports overall F-scores of 0.802 (news) and 0.735 (ASR) but without per-language breakdowns; the Hugging Face model card `verbit/hebrew_punctuation` releases an AlephBERT-based fine-tuned system without an accompanying benchmark. We therefore include Li and Lin’s aggregate numbers as the closest available Hebrew/Russian reference point in Table 7 and explicitly mark the entry as multilingual / not directly comparable. Building directly comparable Kyrgyz–Hebrew–Arabic–Russian punctuation benchmarks (with matched domains, splits, and label schemes) would, on its own, constitute an interesting follow-up cross-lingual study, and we hope the present Kyrgyz dataset and models can serve as one anchor point for such future work.

5.4 ASR Integration

To demonstrate that state-of-the-art ASR systems do not themselves produce the punctuation required for a reader-usable Kyrgyz transcript, we evaluate our post-processor on top of two candidate Kyrgyz ASR systems using 200 audio samples drawn at random (seed 42) from the Kyrgyz subset of Mozilla Common Voice.

Whisper-large-v3 does not support Kyrgyz. OpenAI’s multilingual Whisper model [20] exposes a fixed inventory of 100 language tokens through its tokenizer³. Of these, seven are Turkic: Turkish (`tr`), Kazakh (`kk`), Uzbek (`uz`), Azerbaijani (`az`), Tatar (`tt`), Bashkir (`ba`), and Turkmen (`tk`). Kyrgyz (`ky`) is *not* among them. Invoking `model.generate(language="ky")`

³<https://github.com/openai/whisper/blob/main/whisper/tokenizer.py>, mirrored in the HuggingFace Transformers library as `transformers.models.whisper.tokenization_whisper.LANGUAGES`.

Table 8. Surface punctuation statistics on 200 Common Voice Kyrgyz utterances: raw ASR output vs. ASR + our XLM-RoBERTa post-processor.

	Whisper-small-kyrgyz	+ our XLM-R
% utterances with any punctuation	88.0	100.0
% utterances ending in .	85.0	100.0
% utterances ending in ?	3.0	0.0
Total commas across 200 utterances	0	589
Total periods across 200 utterances	170	200

raises `ValueError: Unsupported language: ky`. Whisper-large-v3 therefore cannot serve as a Kyrgyz ASR baseline out of the box. NVIDIA’s Parakeet family, likewise, does not have a Kyrgyz-capable checkpoint at the time of writing.

Kyrgyz-fine-tuned Whisper. We therefore use `UlutSoftLLC/whisper-small-kyrgyz` [22], a Whisper-small checkpoint fine-tuned on Kyrgyz speech. On our 200-utterance Common Voice sample it achieves WER 0.503 and CER 0.289 (computed after the standard Common Voice normalization, i.e. lowercasing and removal of punctuation).

Punctuation produced by ASR vs. our post-processor. Table 8 compares the surface punctuation of the raw ASR output with the same output passed through our best model (XLM-RoBERTa-base fine-tuned on the augmented training set). Existing punctuation is stripped from the ASR output before the post-processor is applied, to avoid double-punctuation.

The most striking observation is the *total absence of commas* from the ASR output: across 200 utterances, Kyrgyz-fine-tuned Whisper inserts zero commas. This mirrors the distribution of its own training data (Common Voice transcripts, which are stripped of punctuation before training). Our post-processor restores commas at an average of 2.9 per utterance and raises sentence-terminal punctuation coverage from 88% to 100%. Question marks are not recovered on this sample because Common Voice Kyrgyz consists almost entirely of declarative sentences; the six question marks predicted by raw Whisper were stripped before post-processing and not re-inserted.

Table 9 shows three representative outputs. The first two illustrate the common case where our post-processor cleanly restores punctuation to a correct ASR transcript. The third illustrates a known limitation: on short, factual utterances, our model—which was trained on literary text where converb clauses are frequent and typically comma-separated—can over-insert commas at intra-clause positions. This domain-shift behavior is discussed further in Section 6.4.

Summary. Modern multilingual ASR systems either do not support Kyrgyz at all (Whisper-large-v3, NVIDIA Parakeet) or produce outputs that lack the comma punctuation required for readability (Kyrgyz-fine-tuned Whisper). Our post-processor is complementary to whichever Kyrgyz ASR is deployed upstream.

5.5 Inference Efficiency

Practical deployment on constrained hardware depends not only on task accuracy but also on inference cost. We benchmarked each fine-tuned model on a single NVIDIA RTX 5080 GPU using the representative Kyrgyz sentence “*Мен мектенке барам ...*” padded to a sequence length of 128 subword tokens. GPU latency is the median of 200 forward passes at batch size 1 (warm-up 10); GPU throughput is measured at batch size 32 over 50 iterations; CPU latency is the median of 20 forward passes at batch size 1. Parameter counts include the 4-way classification head.

Relative to XLM-R (the accuracy winner), KyrgyzBERT is $7.8\times$ smaller in parameters and disk footprint, $2.9\times$ faster in single-sample GPU latency, $4.3\times$ faster in GPU throughput, and $3.7\times$ faster in single-sample CPU latency. Although

Table 9. Representative outputs from the ASR + post-processing pipeline on Common Voice Kyrgyz. Rows 1–2 are typical successful cases; row 3 is an intentionally selected failure case showing intra-clause comma over-insertion on a short factual utterance, included for transparency and discussed in Section 6.4.

Stage	Text
Reference (CV, normalized)	алар президентке баш ийишет
Whisper-small-kyrgyz + our XLM-R	Алар президентке баш ийишет. Алар президентке баш ийишет.
Reference (CV, normalized)	анткени алар таза технологияларды колдонушат
Whisper-small-kyrgyz + our XLM-R	Анткени алар таза технологияларды колдонушат. Анткени алар таза технологияларды колдонушат.
Reference (CV, normalized)	төрт кишинин курман болгону дароо ырасталды
Whisper-small-kyrgyz + our XLM-R	Төрт кишинин курман болгону дароо ырасталды. <i>Төрт кишинин, курман, болгону дароо ырасталды.</i> (over-inserted commas)

Table 10. Inference efficiency for the three fine-tuned models (single NVIDIA RTX 5080; sequence length 128).

Model	Params (M)	Disk (MB)	GPU latency (ms, bs=1)	GPU throughput (samples/s, bs=32)	CPU latency (ms, bs=1)
mBERT	177.3	2029	3.69	1,032	35.2
XLM-RoBERTa	277.5	3175	3.71	1,043	35.0
KyrgyzBERT	35.6	407	1.29	4,495	9.5

KyrgyzBERT loses 2.3 points of weighted F1 on the main-results table, its efficiency advantage makes it the preferable choice for real-time ASR post-processing on resource-constrained devices. mBERT occupies an intermediate point with accuracy between XLM-R and KyrgyzBERT but no clear efficiency advantage over XLM-R.

6 Error Analysis and Discussion

6.1 Comma Prediction Challenges

Comma prediction remains the most challenging class for all three models, with XLM-RoBERTa achieving F1 0.767, mBERT 0.698, and KyrgyzBERT 0.660. The difficulty is linked to Kyrgyz converb constructions. Converb clauses (marked by suffixes such as *-p*, *-yp*, *-ip*) function as subordinate clauses and are typically separated by commas. However, the models frequently fail to recognize these morphological cues as comma indicators, particularly in shorter clauses where the converb suffix is the only signal. For example, in “*mushtumdaryn tүйүp, kayra achty*” (‘clenching his fists, he opened [them] again’), the converb suffix *-p* signals a comma, but the model may not identify it as a clause boundary in the absence of other syntactic cues. XLM-RoBERTa’s advantage over mBERT (+0.069 F1) and over KyrgyzBERT (+0.107 F1) suggests that its larger and more diverse pre-training corpus better captures the distributional patterns around Kyrgyz clause boundaries. The ASR-integration study in Section 5.4 further exposes the complementary weakness: the model is trained on literary text and therefore tends to *over-insert* commas on short, factual utterances from Common Voice, where converb-based subordinate clauses are rare.

Table 11. QUESTION F1 before and after augmentation. “Original” is the model trained on the 14,028-sentence corpus evaluated on its 125-question test split; “Augmented” is the model trained on the 16,028-sentence corpus evaluated on its 470-question test split.

Model	Original	Augmented	Δ
mBERT	0.643	0.809	+0.166
XLM-RoBERTa	0.704	0.830	+0.126
KyrgyzBERT	0.410	0.738	+0.328

6.2 Question Mark Detection and the Effect of Augmentation

In the initial 14,028-sentence dataset, the QUESTION class was sharply under-represented: only 940 tokens in total, 125 in the test set, and F1-scores of 0.704 (XLM-R), 0.643 (mBERT), and 0.410 (KyrgyzBERT). We therefore expanded the training data with 2,000 additional question-bearing sentences (Section 4.2.1), tripling the QUESTION token count and quadrupling its test-set support (125 $\rightarrow$ 470). Table 11 shows the effect of the expansion on QUESTION F1 for each model.

All three models benefit substantially, and the largest absolute gain (+0.328 F1) is observed for KyrgyzBERT—the model with the smallest pre-training corpus. This is consistent with the interpretation that the residual difficulty on QUESTION is primarily a data-imbalance problem rather than a model-capacity or architecture problem: once sufficient training support is available, even a 35.6M-parameter monolingual model closes most of the gap to XLM-RoBERTa. Kyrgyz forms questions through suffixal particles (*-by/-py*) and intonation rather than word-order inversion, which makes question detection a morphological signal-detection task that responds well to additional examples. Focal loss [12] and class-weighted training are complementary techniques that may further improve question detection in future work.

6.3 Subword Tokenization Effects

All three models fragment agglutinative Kyrgyz words into multiple subword tokens, but with different tokenization algorithms and training data: mBERT uses WordPiece trained primarily on Wikipedia; XLM-RoBERTa uses SentencePiece BPE trained on a larger and more diverse CommonCrawl corpus that includes Kyrgyz text; and KyrgyzBERT uses WordPiece trained exclusively on Kyrgyz. In every case, the punctuation label is attached to the last subword of each word, so the model must propagate information across subword boundaries. This is particularly challenging for long Kyrgyz words with multiple suffixes, where the punctuation-relevant morphological information (for example, a converb marker) may be several subword positions away from the token that carries the punctuation label. XLM-RoBERTa’s consistent advantage across classes suggests that its tokenizer better preserves the morphological structure of Kyrgyz words; KyrgyzBERT, despite a purely Kyrgyz vocabulary, is limited by its smaller pre-training corpus and shallower encoder.

6.4 Limitations

This study has several limitations that scope its conclusions and point to concrete avenues for future work.

Domain. Our dataset consists primarily of formal written text from literary works, Wikipedia, and news portals. Performance on conversational Kyrgyz or on truly spontaneous ASR-generated text may differ substantially. The ASR-integration experiment in Section 5.4 partly addresses this gap by evaluating on Common Voice audio, but a large-scale conversational benchmark for Kyrgyz does not yet exist.

Punctuation coverage. As detailed in Section 2.3, we model three punctuation marks (., ,, ?) that together account for 98.6% of non-paired punctuation tokens in our corpus. Paired marks (quotation marks, parentheses), the dash (–) used

as a copula substitute, and low-frequency marks (:, ;, !, ...) are explicitly excluded for the reasons given there. Extending coverage to these would require, respectively, structured decoders for paired marks, a Kyrgyz dependency parser for dash placement, and targeted data collection for the rare classes.

Data-augmentation scope. The question-mark expansion described in Section 4.2.1 sources additional sentences from the same corpus family (literary and news Kyrgyz). It therefore increases support for the QUESTION class without broadening the genre distribution. A natural extension would be to draw interrogative sentences from conversational sources (e.g. subtitle or dialogue corpora) once these become available.

Domain transfer to ASR output. The ASR integration study (Section 5.4) shows that our model, trained on literary text, can *over-insert* commas on the shorter and more factual utterances of Common Voice. On this out-of-domain input the model’s literary-trained comma prior does not match the local discourse structure. A deployment pipeline would benefit either from a domain-adapted comma classifier or from a simple confidence-threshold mechanism at inference; both are left for future work.

ASR integration evaluation. Common Voice transcripts are normalized (lowercased, no punctuation), so the ASR output cannot be compared to a ground-truth punctuated reference. We therefore report only surface punctuation statistics (% of utterances with a given mark). A Kyrgyz audio corpus with ground-truth punctuation would enable direct end-to-end punctuation-F1 evaluation of the ASR+post-processor pipeline; constructing such a corpus is itself a contribution we leave for future work.

Cross-Turkic transfer. Although we situate Kyrgyz relative to Turkish and other Turkic languages, our experiments do not quantify cross-Turkic transfer (e.g. training on Turkish punctuation and evaluating on Kyrgyz). Given the sharp differences in script (Cyrillic vs. Latin) and available resources, we expect moderate transfer only, but a systematic study would strengthen the broader Turkic-NLP literature.

7 Conclusion

We presented the first punctuation restoration system for Kyrgyz, comparing three transformer models—mBERT, XLM-RoBERTa-base, and the Kyrgyz-specific monolingual KyrgyzBERT—fine-tuned on an augmented dataset of 16,028 sentences. XLM-RoBERTa-base achieves the best performance with a weighted F1-score of 0.944, outperforming mBERT (0.930), KyrgyzBERT (0.921), and a rule-based baseline (0.768). Contrary to a simple *specialization-beats-generalization* expectation, the Kyrgyz-specific KyrgyzBERT—although attractive from an inference-efficiency perspective ($7.8\times$ fewer parameters, up to $4.3\times$ faster inference)—does not surpass XLM-RoBERTa on this task; the pre-training scale advantage of the multilingual model dominates.

Per-class analysis confirms that period restoration is near-perfect (F1 0.97) across all models, while comma prediction remains the most challenging class (F1 0.767 for XLM-R) due to the interaction between Kyrgyz converb constructions and punctuation conventions. A targeted expansion of question-bearing sentences raises QUESTION-class F1 by 0.13 (XLM-R) to 0.33 (KyrgyzBERT), illustrating that the remaining gap on rare-class performance is primarily a data-imbalance problem. An ASR-integration experiment on Common Voice Kyrgyz shows that even a Kyrgyz-fine-tuned Whisper produces zero commas across 200 utterances, while our post-processor brings sentence-terminal coverage to 100% and adds 589 commas.

Future work will focus on: (1) a Kyrgyz audio corpus with ground-truth punctuation to enable direct end-to-end evaluation of the ASR+post-processor pipeline, (2) domain adaptation or confidence thresholding to mitigate comma

over-insertion on short ASR utterances, (3) cross-Turkic transfer learning using Turkish, Kazakh, and Uzbek data, (4) incorporating morphological features to improve comma prediction at converb boundaries, (5) extending coverage to paired and rare punctuation marks via structured decoders and focal loss, and (6) integrating prosodic features for joint ASR–punctuation models.

Acknowledgments

During the preparation of this manuscript, we made limited use of large language models (LLMs), specifically Anthropic’s Claude, to assist with language refinement and organization of some sections. All technical content, experimental design, dataset construction, model training, and analysis were developed entirely by the authors. The LLMs were not used for ideation of methods, data analysis, or generation of results.

References

- [1] Tanvirul Alam, Akib Khan, and Firoj Alam. 2020. Punctuation Restoration using Transformer Models for High- and Low-Resource Languages. In *Proceedings of the Sixth Workshop on Noisy User-generated Text (W-NUT), EMNLP*. 132–142.
- [2] Anton Alekseev and Timur Turatali. 2024. KyrgyzNLP: Challenges, Progress, and Future. In *Proceedings of the 23rd International Conference on Analysis of Images, Social Networks and Texts (AIST 2024), Lecture Notes in Computer Science*, vol. 15419. Springer.
- [3] Vipul Chordia. 2021. PunKtuator: A Multilingual Punctuation Restoration System for Spoken and Written Text. In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations (EACL)*. 312–320.
- [4] Tilek Chubakov. 2026. KyrText: A Multi-Domain Large-Scale Corpus for Kyrgyz Language. In *Proceedings of the Second Workshop on Language Models for Low-Resource Languages (LoResLM 2026)*. <https://aclanthology.org/2026.loreslm-1.39>
- [5] Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. 2020. Unsupervised Cross-lingual Representation Learning at Scale. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*. 8440–8451.
- [6] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*. 4171–4186.
- [7] Xue-Yong Fu, Cheng Chen, Md Tahmid Rahman Laskar, Shashi Balasubramanian, Tista Nath Bhatt, and Simon Corston-Oliver. 2021. Improving Punctuation Restoration for Speech Transcripts via External Data. In *Proceedings of the Seventh Workshop on Noisy User-generated Text (W-NUT), EMNLP*.
- [8] Yasin Görmez, Halil Arslan, and Mustafa Elyakan. 2025. Developing Deep Learning Models for Turkish Automatic Punctuation Restoration Using a Novel Dataset. *ACM Transactions on Asian and Low-Resource Language Information Processing* 24, 11 (2025), 1–15. doi:10.1145/3772087
- [9] Nuno M. Guerreiro, Ricardo Rei, and Fernando Batista. 2021. Towards Better Subtitles: A Multilingual Approach for Punctuation Restoration of Speech Transcripts. *Expert Systems with Applications* 186 (2021), 115747. doi:10.1016/j.eswa.2021.115747
- [10] Uygur Kurt and Aykut Çayır. 2023. Transformer Based Punctuation Restoration for Turkish. In *Proceedings of the 8th International Conference on Computer Science and Engineering (UBMK)*. IEEE, 169–174. doi:10.1109/UBMK59864.2023.10286690
- [11] Xinjian Li and Edward Lin. 2020. A 43-Language Multilingual Punctuation Prediction Neural Network Model. In *Proceedings of Interspeech*. 1067–1071. doi:10.21437/Interspeech.2020-1250
- [12] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. 2017. Focal Loss for Dense Object Detection. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*. 2980–2988.
- [13] Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. 2019. RoBERTa: A Robustly Optimized BERT Pretraining Approach. *CoRR* abs/1907.11692 (2019).
- [14] Adilet Metinov. 2026. Ensemble Methods for Low-Resource Russian-Kyrgyz Machine Translation: When Diverse Models Beat Better Models. In *Proceedings of the Workshop on Low-Resource Machine Translation (LoResMT 2026)*. <https://aclanthology.org/2026.loresmt-1.24>
- [15] Adilet Metinov, Gulida M. Kudakeeva, and Gulnara D. Kabaeva. 2025. KyrgyzBERT: A Compact, Efficient Language Model for Kyrgyz NLP. *arXiv preprint arXiv:2511.20182* (2025).
- [16] Attila Nagy, Bence Bial, and Judit Ács. 2021. Automatic Punctuation Restoration with BERT Models. *arXiv preprint arXiv:2101.07343* (2021).
- [17] Binh Nguyen, Vu Bao Hung Nguyen, Hien Nguyen, Pham Ngoc Phuong, The-Loc Nguyen, Quoc Truong Do, and Luong Chi Mai. 2019. Fast and Accurate Capitalization and Punctuation for Automatic Speech Recognition Using Transformer and Chunk Merging. In *22nd Conference of the Oriental COCOSDA (O-COCOSDA)*. 1–5.
- [18] Yitong Pan, Dushyant Chauhan, Pooja Ramesh, and Narendar Gupta. 2022. Punctuation Restoration for Multilingual ASR Outputs Using Pre-trained Language Models. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*. 4354–4363.

- [19] Martin Poláček, Petr Červa, Jindřich Žďánský, and Lenka Weingartová. 2023. Online Punctuation Restoration using ELECTRA Model for Streaming ASR Systems. In *Proceedings of Interspeech*. 446–450. doi:10.21437/Interspeech.2023-664
- [20] Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. 2023. Robust Speech Recognition via Large-Scale Weak Supervision. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*. 28492–28518.
- [21] Abdelrahman Sakr and Marwan Torki. 2023. AraPunc: Arabic Punctuation Restoration Using Transformers. In *Proceedings of the 20th ACS/IEEE International Conference on Computer Systems and Applications (AICCSA)*. IEEE, 1–6. doi:10.1109/AICCSA59173.2023.10479326
- [22] UlutSoft LLC. 2024. whisper-small-kyrgyz: a Kyrgyz-fine-tuned Whisper model. <https://huggingface.co/UlutSoftLLC/whisper-small-kyrgyz>.
- [23] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. 2017. Attention is All You Need. In *Advances in Neural Information Processing Systems 30 (NeurIPS)*. 5998–6008.
- [24] Jason Wei and Kai Zou. 2019. EDA: Easy Data Augmentation Techniques for Boosting Performance on Text Classification Tasks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.
- [25] Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, Tim Rault, Rémi Louf, Morgan Funtowicz, and Jamie Brew. 2019. HuggingFace’s Transformers: State-of-the-Art Natural Language Processing. *arXiv preprint arXiv:1910.03771* (2019).
- [26] Jiangyan Yi and Jianhua Tao. 2019. Self-Attention Based Model for Punctuation Prediction Using Word and Speech Embeddings. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 7270–7274.

Response to Reviewers

Paper: Punctuation Restoration for Kyrgyz Language: A Comparative Study of Multilingual Transformer Models

Manuscript ID: TALLIP-26-0124

Authors: (omitted for anonymous review)

Dear Associate Editor and Reviewers,

We thank the Associate Editor and all three reviewers for their careful reading and constructive feedback. Reviewer 1 recommended *Minor Revision*, Reviewer 2 recommended *Accept*, and Reviewer 3 together with the Associate Editor recommended *Major Revision* citing four specific concerns. We have addressed every concern in this revision. The main additions are:

- (1) **KyrgyzBERT evaluation** (new rows in Tables 5 and 6; discussion in Section 5.1).
- (2) **Question-mark data augmentation** (new Section 4.2.1; all three models re-trained on the expanded dataset; Table 11 shows substantial gains on the QUESTION class).
- (3) **ASR integration experiment** with Whisper-large-v3 and Whisper-small-kyrgyz (new Section 5.4).
- (4) **Complete Kyrgyz punctuation inventory** and explicit justification of scope (new Section 2.3; revised Limitations, Section 6.4).
- (5) **Inference-efficiency benchmark** of all three models (new Section 5.5).
- (6) **Extended cross-lingual comparison** with Arabic and a transparent discussion of the per-language F1 gap for Hebrew and Russian (Table 7, Section 5.3).

All new experiments, checkpoints, dataset splits, and figures are reproducible from our public release: a GitHub repository (code, results, figures), a Hugging Face dataset repository (original + augmented + sampled question-bearing splits), and three Hugging Face model repositories (XLM-RoBERTa, mBERT, and KyrgyzBERT fine-tuned checkpoints). The exact URLs and DOIs are withheld in this anonymous-review version and will be provided in the camera-ready manuscript.

Reviewer comments are quoted below verbatim; our responses reference the specific sections, tables, and files added in the revision.

Associate Editor

“Missing mentioning and ideally using KyrgyzBert (<https://huggingface.co/metinovadilet/KyrgyzBert>), e.g., fine-tuning this model vs the Roberta variant. Fine-tuning a model pre-trained on the language is presumably a better and more convincing way to achieve the outcome.”

Response. We have fine-tuned KyrgyzBERT [15] with the same hyper-parameters, split, and seed as mBERT and XLM-RoBERTa. Results are now reported in Tables 5 and 6 and discussed in Section 5.1. On the augmented test set KyrgyzBERT achieves weighted F1 0.921, *below* both XLM-R (0.944) and mBERT (0.930). Per-class F1 on the rare QUESTION class is 0.738 for KyrgyzBERT vs. 0.830 for XLM-R. We discuss the finding explicitly: despite being pre-trained exclusively on Kyrgyz, KyrgyzBERT has fewer parameters (35.6 M vs. 277 M for XLM-R) and a smaller pre-training corpus, and in this task the multilingual pre-training scale outweighs monolingual specialization. We also analyze KyrgyzBERT’s trade-off in the new inference-efficiency section (Section 5.5), where it is $7.8\times$ smaller, $2.9\times$ faster on GPU, and $3.7\times$ faster on CPU than XLM-R — a useful property for on-device ASR post-processing.

“[M]issing relevant baselines: for example, Show SOTA Whisper ASR model and parakeet from NVIDIA failing in the task of properly doing annotations and then show how capable your trained model is.”

Response. Addressed in the new Section 5.4. Two findings are documented: (i) Whisper-large-v3’s tokenizer lists exactly 100 languages, seven of them Turkic (tr, kk, uz, az, tt, ba, tk), but ky is not among them; `model.generate(language="ky")` raises `ValueError: Unsupported language: ky`. Whisper-large-v3 therefore cannot serve as a Kyrgyz ASR baseline out of the box. (ii) NVIDIA Parakeet is currently released only in English-only or 25-language multilingual variants; none include Kyrgyz.

In place of Whisper-large-v3 we evaluate `UlutSoftLLC/whisper-small-kyrgyz` [22], a Kyrgyz-fine-tuned checkpoint, on 200 Common Voice utterances. Raw output ends 85% of utterances with a period, 3% with a question mark, and contains *zero* commas. Passing the same output through our XLM-R post-processor raises sentence-terminal coverage to 100% and adds 589 commas (mean 2.9 per utterance). Section 5.4 also reports a side-finding: the model over-inserts commas on short declarative utterances, which we discuss as a domain-transfer limitation in Section 6.4.

“Only very few punctuation marks are addressed. This work would be stronger if all punctuation is treated, or the authors defended their decision not to.”

Response. The new Section 2.3 (“Punctuation Inventory of Written Kyrgyz”) lists every punctuation mark attested in our corpus — 12 marks in total — together with frequency, function, and a yes/no flag for model coverage (Table 1). We then explicitly justify the exclusion of each unmodeled mark: ! is almost never sentence-final in written Kyrgyz; the dash (–) is a copula substitute whose placement depends on the syntactic parse; «», “”, () are paired marks that token-level classification cannot balance; :, ;, -, ... each constitute <1% of all punctuation tokens. The three marks we do model (., ,, ?) jointly cover 100% of sentence-final positions and 98.6% of non-paired punctuation tokens. Section 6.4 has been revised to cross-reference this inventory and name each excluded mark.

“[M]issing handling of rare punctuation. [...] Consider synthesizing more question mark sentences and adding them to your training data, to improve the question mark classification.”

Response. The new Section 4.2.1 describes the addition of 2,000 question-bearing sentences drawn from an independent in-house cleaned Kyrgyz text corpus (a 1.6M-line plain-text collection curated by the authors during prior data-collection work, with the same literary/news/Wikipedia domain mix as the primary dataset, de-duplicated against the original training set; the sampled question-bearing subset is released as `question_data.json`). The QUESTION token count rises from 940 to 2,938 (3.1×) and the test-set support from 125 to 470 (3.8×). All three models were re-trained on the augmented dataset with identical hyper-parameters. The effect on QUESTION F1 is large for every model (Table 11): +0.166 for mBERT, +0.126 for XLM-R, and +0.328 for KyrgyzBERT. The augmented-data versions are used as the main results throughout the revised manuscript.

Reviewer 1 (Minor Revision)

“I think this is a valuable contribution to resources for Kyrgyz. Although the approach is not novel, the dataset should be useful to researchers working on the Kyrgyz language.”

We thank the reviewer for the positive assessment. The revised manuscript has undergone a copy-editing pass per this reviewer’s recommendation. No other specific changes were requested.

Reviewer 2 (Accept)

“The datasets statistics are showing the 840 question marks and minus the held out validation/test set it’ll get even lesser. [...] There’s a simple hack of synthesizing more question mark sentences and adding them to your training data, that should improve the question mark classification class.”

Response. We thank the reviewer for this specific, actionable suggestion. A small clarification on the dataset statistics first: the original submission reports 940 QUESTION tokens (Table 3), and we believe “840” in the review is a minor transcription typo for 940 — the broader point that the class is severely under-represented is fully correct. As detailed in our response to the Associate Editor above, we have added 2,000 additional question-bearing sentences to the training set; the QUESTION token count rises from 940 to 2,938 ($3.1\times$), the test-set support from 125 to 470 ($3.8\times$), and the QUESTION F1 improves by 0.126 points for XLM-R and 0.328 points for KyrgyzBERT. See Section 4.2.1 and Table 11.

Reviewer 3 (Major Revision)

“1. Update the baseline to have Arabic, Hebrew and (if possible Russian)”

Response. We respectfully note that the Associate Editor explicitly accepted a Kyrgyz-only experimental scope in the decision letter: “*if the authors state they focus on Kyrgyz, since this is what they have spent the time pre-processing, I think it’s fine to stick with only Kyrgyz, without expanding to other languages.*” The revised manuscript adopts this scope: the Title, Abstract, and Introduction now state explicitly that the contribution is the first Kyrgyz punctuation-restoration system, and we do not run new fine-tuning experiments on Hebrew or Russian. We nonetheless extended the cross-lingual comparison (Table 7) using published numbers, as follows:

- **Arabic:** we added the result of Sakr and Torki [21], whose XLM-RoBERTa model achieves macro F1 0.785 on the AraPunc corpus. New row in Table 7.
- **Hebrew:** after a thorough literature search we did not find a peer-reviewed publication reporting a per-language punctuation-restoration F1 for Hebrew. The `verbit/hebrew_punctuation` model card referenced in the review contains no F1 score, paper, or blog post. The closest public reference is Li and Lin [11], whose 43-language model (including Hebrew) reports aggregate F-scores of 0.802 (news) / 0.735 (ASR) without per-language breakdowns. We cite this paper as a multilingual-baseline row in Table 7.
- **Russian:** the same situation holds. Li and Lin [11] includes Russian in the 43-language bundle but does not publish per-language F1.

We transparently note these gaps in the text accompanying Table 7.

“2. Show SOTA Whisper ASR model and parakeet from NVIDIA failing in the task of properly doing annotations and then show how capable your trained model is.”

Response. Addressed in Section 5.4 (see response to the Associate Editor). Whisper-large-v3 does not support Kyrgyz; NVIDIA Parakeet has no Kyrgyz-capable variant. Using `UlutSoftLLC/whisper-small-kyrgyz` [22] as the strongest available Kyrgyz ASR, we show that it produces zero commas across 200 Common Voice utterances, and that our post-processor raises sentence-terminal coverage from 88% to 100% and adds 589 commas.

“3. Use Kyrgyz specific pre-trained model that I have specified to show how fine-tuning that + maybe zero shot be valuable than fine-tuning an outdated model like roberta variant.”

Response. We have evaluated KyrgyzBERT as requested. Empirically, KyrgyzBERT does *not* outperform XLM-RoBERTa in our setting: overall weighted F1 0.921 vs. 0.944, and QUESTION F1 0.738 vs. 0.830. We attribute this to the substantial gap in pre-training data volume: XLM-RoBERTa was trained on 2.5 TB of filtered CommonCrawl covering 100 languages including Kyrgyz, whereas KyrgyzBERT is a compact 6-layer / 35.6 M-parameter model pre-trained on a smaller Kyrgyz-only corpus. Section 5.1 presents this as an honest empirical finding: monolingual specialization is not currently sufficient to overcome the pre-training scale advantage of multilingual XLM-R for Kyrgyz punctuation restoration. We

do, however, show (Section 5.5) that KyrgyzBERT has a $7.8\times$ parameter advantage and a $2.9\text{--}3.7\times$ latency advantage, which makes it an attractive option for resource-constrained deployment.

We also performed the zero-shot evaluation the reviewer suggested via MLM mask-filling (Section 5.1, paragraph “Zero-shot KyrgyzBERT”). Result: weighted F1 0.769, with PERIOD F1 0.001 and QUESTION F1 0.000; the zero-shot model essentially never predicts sentence-terminal punctuation because the pre-training distribution is dominated by commas. This empirically confirms that fine-tuning on labeled punctuation data is necessary even for a Kyrgyz-specific MLM.

“4. Please, outline all the punctuations present for Kyrgyz and how many you are addressing through your contribution. And update limitations accordingly.”

Response. The new Section 2.3 provides the complete Kyrgyz punctuation inventory with per-mark frequency, function, and coverage flag (Table 1). Section 6.4 has been rewritten to cross-reference this inventory and name every excluded mark with its reason.

“Including how hardware efficient their model is in terms of inference.”

Response. Addressed in the new Section 5.5 and Table 10: parameters, disk footprint, GPU single-sample latency (bs=1), GPU throughput (bs=32, samples/s), and CPU single-sample latency for all three fine-tuned models, measured on an NVIDIA RTX 5080.

“Quality of English: Heavy copy-editing.”

Response. The entire manuscript has undergone a comprehensive copy-editing pass that addresses both this reviewer’s request for heavy copy-editing and Reviewer 1’s request for a light pass. Specifically, we (i) unified spelling to American English throughout (e.g. *tokenization*, *labeled*, *recognize*, *modeled*); (ii) tightened phrasing in the Abstract, Introduction, and Conclusion; (iii) rewrote the Limitations section (Section 6.4) into six focused paragraphs; (iv) added explicit topic sentences and cross-references to every new table and figure; and (v) corrected typographical and grammatical issues found during the pass. We hope the revised text reads more cleanly than the original submission.

We are grateful for the reviewers’ time and the quality of the suggestions, which have measurably strengthened the paper.

Sincerely,
The authors